

Network Security & Wireless Networks



Wireless Networks

Firewalls

IDS / IPS

CIA Triad

Best Practices

What is Network Security?

Protecting everything — data, devices, systems, and users

Network security is a set of practices, policies, and tools designed to protect all digital assets within a network from unauthorized access, damage, or misuse.

Think of it like a digital fortress — every layer must be secured:

- 🔒 Who can enter? (Access Control)
- 📦 What can they take? (Data Protection)
- 🚨 Who alerts us if something's wrong? (Monitoring)

C

Confidentiality

Only authorized users can see sensitive data. Example: Your password is encrypted — even the database admin can't read it.

I

Integrity

Data stays accurate and untampered. Example: A hacker modifies a bank transfer record — integrity controls detect the change.

A

Availability

Systems are accessible when needed. Example: A DDoS attack floods a server — availability is the first casualty.

Wireless Networks

Connecting devices without cables — using radio waves



✓ ADVANTAGES

- Mobility** — Move freely within coverage range
- Easy Setup** — No drilling or cable runs needed
- Scalable** — Add devices with zero physical changes
- Real World:** Your home Wi-Fi connects phones, TVs, laptops all wirelessly

⚠ DISADVANTAGES

- Interference** — Walls, microwaves, Bluetooth disrupt signal
- Security Risk** — Signals can be intercepted without encryption
- Slower Speed** — Wired is faster for same-generation hardware
- Attack Scenario:** A hacker sets up an 'Evil Twin' AP to steal your credentials

KEY FREQUENCY BANDS

2.4 GHz

- Older Wi-Fi (802.11b/g/n)
- ✓ Better wall penetration
 - ⚠ More interference

5 GHz

- Newer Wi-Fi (802.11ac/ax)
- ✓ Faster speeds
 - ⚠ Shorter range

Cellular

- 4G LTE / 5G
- ✓ Wide area coverage
 - ⚠ Requires tower infra

Key Wireless Devices

Router · Mobile Hotspot · Cell Tower — how each connects you



Wireless Router

- Acts as traffic director — routes data to correct destination
- Combines routing + Wi-Fi access point in one box
- Has WAN port (to internet), LAN ports (wired devices), antennas

📌 Real World: Your home router connects to ISP via WAN and shares Wi-Fi to all your devices via LAN/Wi-Fi.

🛡️ Security: Default router passwords are a major vulnerability — always change them!



Mobile Hotspot

- Turns phone's cellular data into a Wi-Fi network others can join
- Uses 4G/5G — not dependent on any physical infrastructure
- Range is small (few meters), and drains battery fast

📌 Real World: On a business trip, share your phone's data connection with your laptop to attend a video call.

🛡️ Security: Always protect your hotspot with a strong password — open hotspots are risky!



Cell Tower (Base Station)

- Large structures housing antennas covering kilometers of area
- BSC (Base Station Controller) manages handoffs between towers
- Connected to core network via fiber or microwave backhaul

📌 Real World: As you drive on a highway, your call doesn't drop because your phone silently switches towers.

🛡️ Security: Rogue cell towers ('IMSI catchers') can intercept calls — a real threat in espionage.

Firewalls



The first line of defense — controlling what enters and leaves your network

A firewall monitors all incoming and outgoing traffic and applies rules to decide: ALLOW or BLOCK. Think of it as a security guard at a building entrance — nobody gets in without passing the rules.

Packet Filtering

OSI Layer 3 & 4

Checks source/dest IP, port, and protocol. Fast but basic — can't inspect data content.

→ Allow port 443 (HTTPS), block everything else.

Stateful Inspection

Layer 4 (Context-aware)

Tracks the full conversation. Only allows replies to outbound requests you made.

→ Blocks inbound TCP if no outbound session exists.

Application Layer (Proxy)

OSI Layer 7

Reads the actual content of traffic. Can block malicious HTTP patterns, not just ports.

→ Web proxy blocks requests with SQL injection strings.

Next-Gen Firewall (NGFW)

Layer 3–7 + Intelligence

Combines everything above + deep packet inspection, threat intel, and app-level control.

→ Blocks ransomware C2 traffic even on port 443.

IDS vs IPS

Intrusion Detection vs Intrusion Prevention — know the difference



IDS — Intrusion Detection System

- 👁️ Watches and monitors all traffic passively
- 🔔 Generates ALERTS when suspicious activity is detected
- ❌ Does NOT block anything automatically
- 📌 Used for: Visibility, forensics, compliance logging
- 📌 **Example:** IDS detects a port scan and logs it, but the scan still completes. SOC analyst investigates.

IPS — Intrusion Prevention System

- 🛡️ Actively intercepts traffic in real time
- 🚫 BLOCKS malicious packets before they reach the target
- ⚡ Inline device — sits directly in the traffic path
- 📌 Used for: Active threat prevention in critical networks
- 📌 **Example:** IPS detects an SQL injection attempt and drops the packet — the attack never hits the database.

DETECTION TECHNIQUES

📁 Signature-Based

Matches traffic against a database of known attack fingerprints. Fast and accurate for known threats — blind to new ones.

📊 Anomaly-Based

Learns 'normal' behavior and flags deviations. Can catch zero-day attacks — higher false positive rate.

💻 Network vs Host

NIDS/NIPS covers the whole network. HIDS/HIPS runs on individual machines (like endpoint antivirus).

Security Best Practices

Layered defense — because no single tool is ever enough

Principle of Least Privilege

Only grant users the minimum access they need to do their job. If a developer doesn't need database admin rights — don't give them.

Used in: IAM, Active Directory, SOC

Regular Updates & Patching

Outdated software is the #1 way hackers get in. The 2017 WannaCry ransomware hit unpatched Windows systems worldwide.

Used in: Patch Management, Blue Team

Monitor & Log Everything

You can't defend what you can't see. Centralized logging (SIEM) lets security teams spot suspicious patterns before damage is done.

Used in: SOC, SIEM, Threat Hunting

Defense in Depth

Layer multiple security tools — Firewall → IPS → Antivirus → EDR. If one layer fails, the next one catches the attack.

Used in: Enterprise Architecture, Red/Blue Teams

Penetration Testing

Hire ethical hackers to find your weaknesses before the bad guys do. Regular pen tests validate your security posture.

Used in: Pentesting, Red Team operations



Key Takeaways

The most important lessons from this module

01

Security = CIA

Every security decision maps back to Confidentiality, Integrity, or Availability. If you're ever unsure why something matters — trace it to the CIA Triad.

02

Wireless = Convenience + Risk

Wi-Fi makes life easy, but radio signals travel through walls and can be intercepted. Encryption (WPA3) and strong passwords are non-negotiable.

03

Firewall ≠ Full Security

A firewall is the gatekeeper — not the whole castle. Knowing which type to use (packet filter vs NGFW) matters in real deployments.

04

IDS Detects, IPS Prevents

Both are essential. Use IDS for forensics and visibility. Use IPS in-line to actively stop threats. Many tools like Suricata can do both.

05

Layers Win

No single tool protects everything. Real security is a layered strategy — firewall + IPS + antivirus + monitoring + training + pen testing.



My Understanding

In my own words — what I actually learned from this module

written in first person — as a student who studied and understood the concepts



Wireless isn't magic — it's radio

Before this module, I just thought Wi-Fi 'happened.' Now I understand there are actual frequency tradeoffs — 2.4 GHz travels farther but gets congested, 5 GHz is fast but short. Choosing the right band is a real engineering decision.



A firewall is only as good as its rules

I always assumed firewalls were magic protection. Now I see they're just rule engines. If you don't define smart rules — like using least-privilege policies — attackers can walk right through. The human defines the logic.



IDS vs IPS changed how I think about defense

Understanding that IDS watches while IPS acts was eye-opening. In a SOC role, you'd use IDS logs to investigate threats. In a production network, IPS protects you in real time. Both are needed — just for different purposes.



Security is never one tool — it's a mindset

The biggest lesson: no single product makes you secure. Real security is layers — firewall, IPS, patching, monitoring, and training people. The weakest link is often not the technology — it's the human using it.